

United Airlines Departure Gate Utilization Optimization at Denver International Airport

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1. Introduction

This project focuses on understanding and improving gate utilization for United Airlines departing flights at Denver International Airport. Denver is United's largest hub, and unlike many carriers that share terminals, United operates with a significant number of fixed gates in Concourse B. Because gate space is limited and costly to expand, airlines often look for ways to optimize how they use these existing gates. Denver is an especially interesting setting for this type of problem because United has unusually high gate ownership at the airport compared to other hubs.

In this project, we analyzed United's July 2025 departure schedule and used optimization techniques to estimate the minimum number of gates needed each day to support all departures without timing conflicts. We calculated gate-occupancy windows for each

departure and determined how often these windows overlap, which allowed us to compute the smallest number of gates required for each day of the month. This gives us a clear picture of how departure-driven gate demand fluctuates over time and how United's schedule shapes daily congestion.

Our initial goal was to build a full optimization model that assigned every United flight to a specific gate. This type of model would have created a complete gate-assignment plan and identified the exact minimum number of gates needed for the entire month. However, the full model became too large for the limits of our Gourbi license, so we shifted to estimating the minimum number of gates per day instead of assigning flights to individual gates.

Even though our approach is simpler than the original plan, it still provides valuable insights. Departure operations consume more predictable time, staff, and resources than arrivals, which makes them the primary drivers of gate occupancy. By focusing on departures and analyzing daily peak overlaps, our model offers a realistic lower bound on the gate capacity United needs. This information can help the airline identify patterns in its schedule, understand where congestion occurs, and explore opportunities to use its gate resources more efficiently.

2. Statement of the Problem and Results

The problem addressed in this project is to determine the minimum number of gates United Airlines needs each day to operate its July 2025 departure schedule at Denver International Airport without gate-time conflicts. Each departure occupies a gate for a period before and after its scheduled departure time, and when several flights fall within overlapping windows, the demand for gates increases.

To measure this daily requirement, we constructed a gate-occupancy interval for every departure and examined each day to find the maximum number of flights occupying gates at the same time. That peak level of simultaneous demand represents the minimum number of gates needed to operate the day's departure schedule safely and without delays. This framework allows us to quantify daily gate needs and understand how the timing of departures shapes United's gate utilization patterns.

Our analysis of United's July 2025 departure schedule shows that the minimum number of gates required per day falls within a narrow range. The full daily results table is included in Appendix 1. Across all thirty-one days, the model indicates that United needs between 48 and 52 gates to operate its departures without any gate-time conflicts under the assumed gate-occupancy windows.

A key finding is that daily gate demand does not depend on the total number of flights in a day but on how those flights are distributed over time. Some days with fewer departures require more gates because many flights are scheduled close together, creating larger

peaks of simultaneous gate use. Conversely, days with more flights may need fewer gates if departures are spread more evenly. A visual comparison of flights and minimum gates is shown in Appendix 3.

We also ran a small sensitivity test to see how changes in the gate-occupancy buffers would affect the results (see Appendix 7,8). The overall pattern did not change, and moderate adjustments led to only small increases in required gates, suggesting that our main findings are reasonably stable.

These results represent a lower bound on gate requirements for departure operations under ideal conditions. While actual operations also involve arrivals, delays, and irregular events that would increase real gate usage, the daily values produced by our model reveal the underlying structure of peak departure demand and provide a useful baseline for understanding gate utilization at Denver.

3. Methodology and Model

Our methodology uses an optimization-based framework to estimate the minimum number of gates United Airlines needs each day to operate its July 2025 departure schedule without timing conflicts. The problem is structured as a capacity-planning analysis in which gate demand is determined by how many flights require access to gates at the same time. This formulation allows us to focus on peak congestion, which is the true driver of gate requirements in airline operations.

We first prepared the data using the Bureau of Transportation Statistics' Detailed Statistics – Departures dataset. For every United flight departing Denver in July 2025, we combined the reported date and scheduled departure time into a single timestamp. We defined a gate-occupancy window for each departure covering 45 minutes before the scheduled departure time and 15 minutes after. This interval approximates the time during which a gate is unavailable due to boarding, loading, fueling, catering, and pushback. This approximation is based on personal experience, not a set industry standard.

These occupancy windows were then mapped to a minute-by-minute timeline for each day. This step converts the raw schedule into a structure that makes it possible to identify when flights overlap and how many gates are needed to accommodate that day's operations. The number of overlapping windows at each minute forms the basis of the constraint in our optimization model.

We used a linear optimization model solved with the Gurobi Optimizer through Python to formalize this process. For each day, the model introduces a single decision variable, , representing the number of gates required (treated as an integer). The objective is to minimize . The constraint enforces that must be at least as large as the number of flights occupying their gate windows at every minute during that day. This ensures the model captures the true peak of simultaneous gate use. We then looped through this process for

each day for the entire month. A short excerpt of the Python implementation is included in Appendix 2. Because the problem is linear and small, it can be solved efficiently while still representing the core logic of gate planning.

This daily capacity-minimization model was chosen over a full flight-to-gate assignment model for two reasons. First, the full assignment model would have included thousands of binary decision variables and conflict constraints, pushing it beyond the limits of a standard academic Gurobi license. Second, for the purpose of determining gate requirements, the minimum number of gates is fully determined by peak simultaneous use rather than by the specific assignment pattern. The simplified model therefore provides the same insight on gate capacity while remaining computationally feasible.

The decision variable, objective, constraints, and assumptions used in the model are listed below for clarity:

Decision Variable: *K = minimum number of gates needed that day (Integer)*

Objective Function: Minimize K

Constraints:

- For each minute t of the day, K ≥ number of flights occupying gates at minute t.
- Non-negativity constraint, K ≥ 0

Assumptions

- United operates roughly 90 gates at DEN
- Each flight occupies a gate from 45 minutes before its scheduled departure until 15 minutes afterward.
- Only departures were modeled

This methodology provides a practical and analytically transparent way to evaluate daily gate requirements without building a full-scale assignment model. It maintains the essential logic of minimizing gate usage while ensuring that no gate is double-booked within any minute of the operational day.

4. Data Collection and Description

The data for this project comes from the Bureau of Transportation Statistics (BTS)

TransStats – Detailed Statistics: Departures dataset. This is a publicly available source

that provides flight-level schedule data for all major U.S. airlines. The dataset is accessible at: https://www.transtats.bts.gov/ONTIME/Departures.aspx (*Detailed Statistics: Departures*, n.d.)

We filtered the dataset to include only United Airlines (UA) flights departing from Denver International Airport (DEN) during July 2025. This gave us the full set of 9,646 scheduled United departures for the month we analyzed.

The fields used from the dataset were:

- FlightDate – the scheduled calendar date of the flight
- Carrier – the airline operating the flight (United = UA)
- Origin – the airport the flight departs from (DEN for all flights in our filtered set)
- Dest – the destination airport for each flight
- CRSDepTime – the scheduled departure time in local time
- CRSArrTime – the scheduled arrival time in local time
- CRSElapsedTime – the published scheduled duration of the flight

These fields allowed us to reconstruct the full departure schedule for each day, create timestamped departure events, and calculate the gate occupancy windows used in our optimization model.

5. Numerical Experiment Results

After preparing the departure data and running the daily optimization model, we obtained minimum gate requirements for all thirty-one days in July. For each day, the model calculated the peak number of overlapping gate-occupancy intervals, which directly corresponds to the minimum number of gates needed to support that day's departure schedule without conflict.

The results show that the number of gates required each day remains very stable throughout the month. A combined bar-and-line visualization of daily flights and daily gate requirements is shown in Appendix 3. Across all days, the minimum gate requirement ranges from 48 to 52 gates, even though the total number of flights varies from day to day. This pattern indicates that peak congestion, more so than total daily volume, determines gate demand. For example, July 5 has fewer departures than many other days but requires the highest number of gates (52) because many of its departures fall within the same time window. Conversely, some days with more departures require fewer gates when the flights are more evenly distributed across the day.

This analysis highlights how United's scheduling structure, particularly its use of departure banks, creates concentrated intervals of gate demand. Weekday averages and comparisons are presented in Appendices 4 and 6, which highlight efficiency differences by day of week. These banks drive the peak overlap patterns that ultimately determine the number of gates required. Even though the total number of flights varies, the clustering of departures produces nearly identical daily gate requirements.

Overall, the results provide a clear lower bound for departure-driven gate needs at Denver. While real operations also involve arrivals, delays, and turnaround variation, these daily values demonstrate the underlying structure of gate usage and offer a consistent baseline against which United can evaluate operational efficiency.

A complete table of daily gate requirements and daily flight counts is included in the Appendix 1.

To test how sensitive our results are to the assumed gate-occupancy window, we varied the pre-departure and post-departure buffers and recomputed the minimum daily gate requirement. The full sensitivity results table is shown in Appendix 7, and the corresponding heatmap visualization appears in Appendix 9. Under our baseline assumptions of 45 minutes before departure and 15 minutes after, the model requires at most 52 gates on the busiest day. When we modestly lengthen either the pre-departure or post-departure buffer to combinations such as (45, 30), (45, 45), (60, 15), or (60, 30), the minimum requirement increases only slightly. With these combinations the maximum requirement increases by just 1, to 53 gates, indicating that the schedule can absorb moderate extensions of gate time without a large increase in peak gate demand. More aggressive buffer expansions, however, have a larger effect. For a window of (60, 45) or (90, 15) the maximum jumps to 69 gates and for the second largest buffer, (90, 30), it

jumps to 84 gates. For the largest buffer, (90, 45), it tops out at 90 gates which is the maximum number of gates United has available for all flights going through Denver, not just departures. These patterns suggest that our baseline estimate of 48-52 gates is robust to reasonable variations in gate-time assumptions, but that conservative buffering, particularly to both pre- and post-departure times can quickly exhaust United's entire supply of gates.

6. Recommendations

Our main quantitative finding is that, under our assumptions, United's scheduled July departures at DEN can be supported with roughly 48–52 gates per day. This stability indicates that, for departures alone, United's gate demand is highly predictable and driven primarily by peak-period clustering rather than overall flight volume. Given that United has access to roughly 90 gates at DEN, the airline operates with a considerable amount of unused gate capacity relative to its departure-driven needs.

Based on these findings, we offer the following recommendations:

1. Use peak-overlap patterns to understand where the schedule is most strained

Our results show that gate demand depends mainly on how departures cluster in certain hours, not on the total number of flights. A practical takeaway is for United to use this kind of peak-overlap analysis when reviewing future schedules. It can help the airline spot the specific time windows that consistently create pressure. Even without changing gates or terminal layout, simply knowing when those peaks occur can guide planning conversations and help United understand how sensitive certain parts of the schedule are.

2. Smooth peak departure banks where feasible

The data show that some days with fewer total flights still require more gates because multiple departures cluster into tight windows. United could benefit from reviewing those peak periods and exploring whether a small number of departure time adjustments could reduce overlap. Even small spacing adjustments can meaningfully decrease gate congestion and improve on-time performance without major schedule changes.

3. Use results to support long-term gate planning and capital decisions

These findings can inform future conversations about gate expansion, modernization, or partnerships at DEN. Since departures alone do not appear to stress United's current gate portfolio, capital expansions are unlikely to be needed for departure operations unless wide-body growth or new operational constraints emerge.

7. Limitations

1. Arrivals were not included. Arrivals can also occupy gates for extended periods, and irregular arrival patterns can increase total gate demand beyond our calculated lower bound.
2. Turnaround variation was not modeled. Different aircraft types require varying ground times, and operational delays were not incorporated.
3. The model assumes exact schedule adherence. Real-world flight operations experience delays, early arrivals, and weather disruptions that increase gate usage.
4. Only one gate-occupancy rule was used. This is a reasonable assumption, but it will not match every situation. Our sensitivity analysis showed that significantly longer gate-use times can quickly increase how many gates are needed, which means the exact buffer matters.

8. Conclusions

Our model gives a straightforward picture of how United's departures use gates at Denver. By focusing on the busiest minutes of each day, we were able to see how clusters of departures, not the total number of flights, drive gate demand. The results suggest that United's departure schedule can be supported with a stable number of gates, which gives the airline some flexibility in how it manages daily operations.

Even though our approach is simplified, it provides a useful starting point. It shows the basic pattern of how gate pressure builds during the day and offers a baseline that United could compare against more detailed operational data. With more information and a stronger optimization setup, this framework could be expanded to include arrivals, real turnaround times, delays, and even full flight-to-gate assignments.

Overall, this project shows how a relatively simple optimization model can still reveal important scheduling patterns and help guide more advanced analysis in the future.

9. Appendix

Appendix 1

Minimum gates required per day:

	Date	NumFlights	MinGates
0	2025-07-01	304	50
1	2025-07-02	311	48
2	2025-07-03	321	49
3	2025-07-04	285	50
4	2025-07-05	282	52
5	2025-07-06	317	49
6	2025-07-07	319	49
7	2025-07-08	304	50
8	2025-07-09	311	48
9	2025-07-10	321	49
10	2025-07-11	319	48
11	2025-07-12	301	50
12	2025-07-13	318	49
13	2025-07-14	319	49
14	2025-07-15	304	50
15	2025-07-16	311	48
16	2025-07-17	321	49
17	2025-07-18	319	48
18	2025-07-19	301	50
19	2025-07-20	318	49
20	2025-07-21	319	49
21	2025-07-22	303	50
22	2025-07-23	310	48
23	2025-07-24	320	49
24	2025-07-25	318	48
25	2025-07-26	301	50
26	2025-07-27	318	49
27	2025-07-28	318	49
28	2025-07-29	303	50
29	2025-07-30	310	48
30	2025-07-31	320	49

Figure 1: Results Table with Minimum Gate Requirement by Day

Appendix 2

```python
#Function for minimum gates for one day
def compute_min_gates_for_day(day_df, buffer_before=45, buffer_after=15):
    """
    Given all flights for a single calendar day in day_df (with column 'dep_dt'),
    compute the minimum number of identical gates required using a tiny Gurobi model:
        min K  s.t.  K >= #flights occupying each minute of the day.
    """

    if day_df.empty:
        return 0

    # Midnight of this specific day
    day_start = pd.Timestamp(day_df["dep_dt"].iloc[0].date())

    # Minutes since midnight for each departure
    t_dep_min = (day_df["dep_dt"] - day_start).dt.total_seconds() / 60.0

    # Gate-occupancy interval per flight: [start_min, end_min)
    start_min = (t_dep_min - buffer_before).astype(int)
    end_min   = (t_dep_min + buffer_after).astype(int)

    # Restrict to the 0..1439 minutes of that calendar day
    T = 24 * 60
    start_min[start_min < 0] = 0
    end_min[end_min > T] = T

    # Build occupancy array: occ[t] = how many flights use a gate at minute t
    occ = [0] * T
    for s, e in zip(start_min, end_min):
        for t in range(s, e):   # interval [s, e)
            occ[t] += 1

    #Gurobi model:
    #subject to K >= occ[t] for all minutes with occ[t] > 0
    m = Model("MinGates_OneDay")
    m.Params.OutputFlag = 0  # silence solver chatter

    K = m.addVar(vtype=GRB.INTEGER, lb=0, name="K")

    for t in range(T):
        if occ[t] > 0:
            m.addConstr(K >= occ[t], name=f"time_{t}")

    m.setObjective(K, GRB.MINIMIZE)
    m.optimize()

    if m.Status == GRB.OPTIMAL:
        return int(K.X)
    else:
        print("Warning: Gurobi status", m.Status, "- using max(occ) instead.")
        return max(occ)


#Loop over all days in July
results = []

unique_days = sorted(flights["day"].unique())
print("Days in data:", unique_days)

for d in unique_days:
    day_df = flights[flights["day"] == d].copy()
    num_flights = len(day_df)

    min_gates = compute_min_gates_for_day(
        day_df,
        buffer_before=BUFFER_BEFORE,
        buffer_after=BUFFER_AFTER,
    )

    results.append({
        "Date": d,
        "NumFlights": num_flights,
        "MinGates": min_gates})

```

Figure 2: Python Code for the Model

Appendix 3

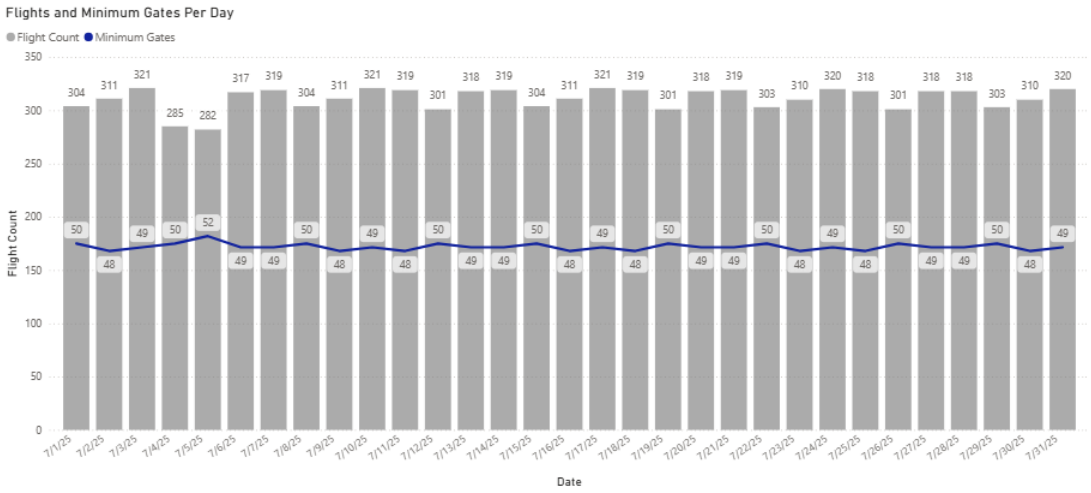


Figure 3: Daily United Flight Counts and Minimum Gate Requirements for July 2025

The figure compares the number of scheduled United departures each day with the minimum number of gates required based on overlapping gate-occupancy windows.

Appendix 4

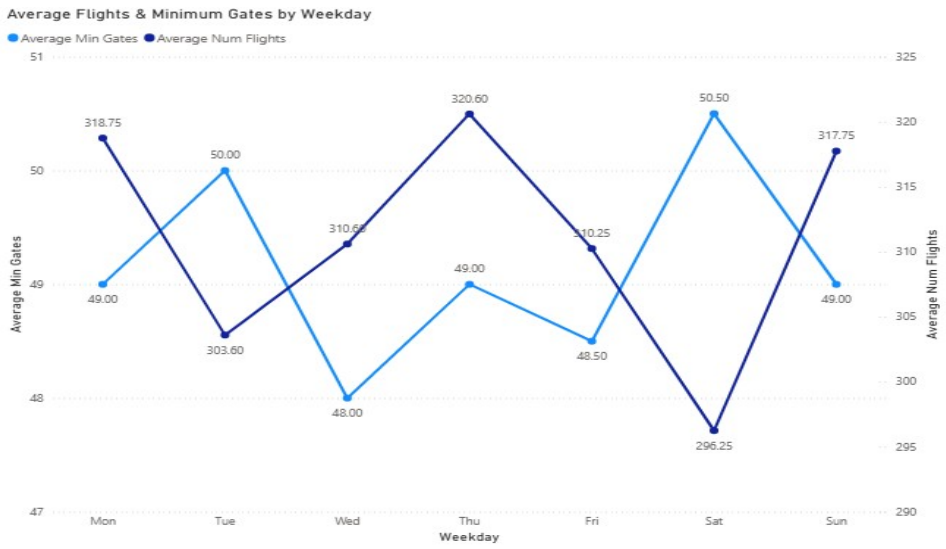
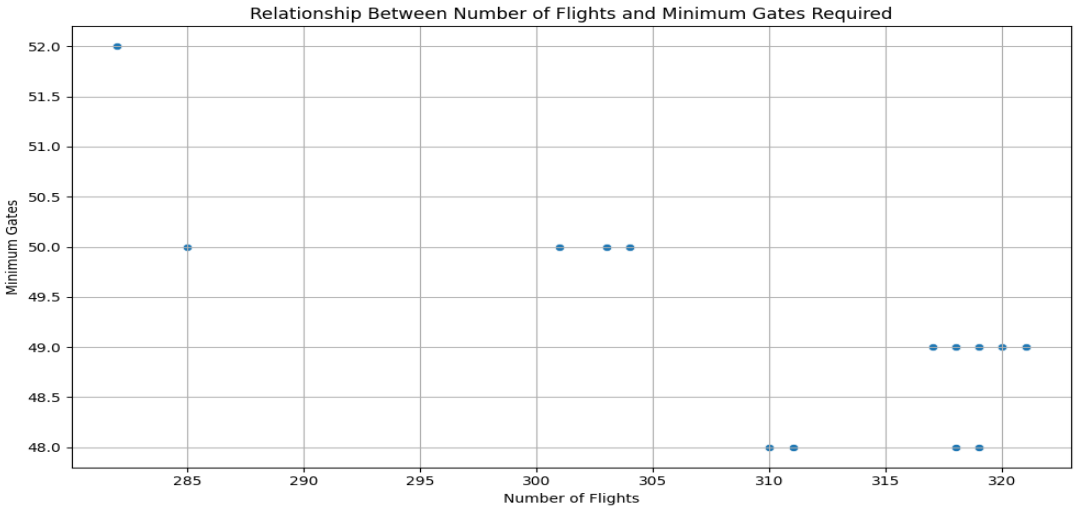


Figure 4: Weekday patterns in flights and gate demand

Visualization of weekday trends in flight counts and minimum gates required, highlighting that certain weekdays use gates less efficiently due to tighter clustering of departures.

Appendix 5



Appendix 6

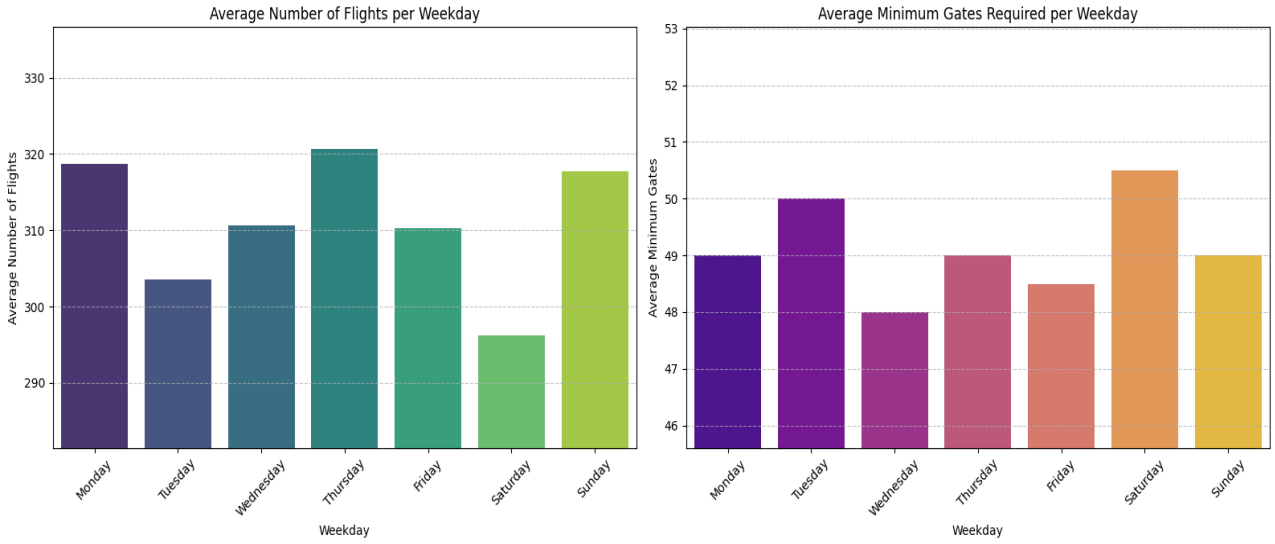


Figure 6: Average number of flights and average minimum gates required by weekday

The plots show that some weekdays need more gates relative to their flight volume, indicating less efficient departure spacing on those days.

Appendix 7

Sensitivity Analysis Results:

	BUFFER_BEFORE	BUFFER_AFTER	Max_Daily_Gates
0	45	15	52
1	45	30	53
2	45	45	53
3	60	15	53
4	60	30	53
5	60	45	69
6	90	15	69
7	90	30	84
8	90	45	90

Figure 7: Sensitivity analysis of maximum daily gate requirements under different gate-occupancy buffer assumptions

The table shows that moderate changes in buffer times have minimal impact on gate needs, but large increases in pre- and post-departure buffers can sharply raise the required number of gates.

Appendix 8

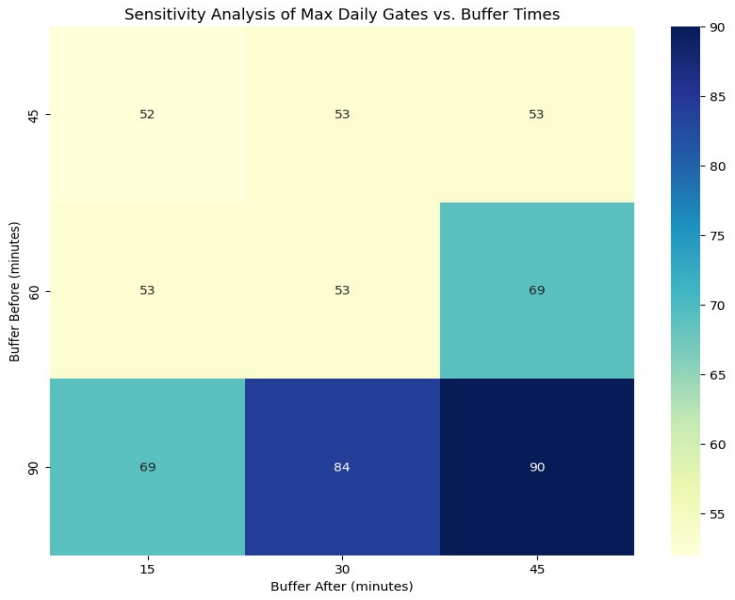


Figure 8: Sensitivity analysis heatmap showing how different buffer times affect the maximum number of gates required

The heatmap illustrates that small increases in buffer times have minimal impact on gate needs, but larger extensions, especially when both buffers increase together, cause gate requirements to rise sharply.

Appendix 9

```python
buffer_before_values = [45, 60, 90]
buffer_after_values = [15, 30, 45]

print(f"BUFFER_BEFORE values for sensitivity analysis: {buffer_before_values}")
print(f"BUFFER_AFTER values for sensitivity analysis: {buffer_after_values}")
```

```
BUFFER_BEFORE values for sensitivity analysis: [45, 60, 90]
BUFFER_AFTER values for sensitivity analysis: [15, 30, 45]
```

```python
sensitivity_results = []

for b_before in buffer_before_values:
    for b_after in buffer_after_values:
        daily_min_gates = []
        for d in unique_days:
            day_df = flights[flights["day"] == d].copy()
            min_gates = compute_min_gates_for_day(
                day_df,
                buffer_before=b_before,
                buffer_after=b_after,
            )
            daily_min_gates.append(min_gates)

        max_daily_gates = max(daily_min_gates)
        sensitivity_results.append({
            "BUFFER_BEFORE": b_before,
            "BUFFER_AFTER": b_after,
            "Max_Daily_Gates": max_daily_gates
        })

sensitivity_df = pd.DataFrame(sensitivity_results)
print("\nSensitivity Analysis Results:")
print(sensitivity_df)
```

Figure 9: Python code used to run the sensitivity analysis on gate-occupancy buffer assumptions

References

ChatGPT (AI Model) [AI-generated content used for editing, coding assistance, and report refinement]. (n.d.). OpenAI ChatGPT. https://chat.openai.com/

Detailed Statistics: Departures. (n.d.). Bureau of Transportation Statistics. https://www.transtats.bts.gov/ONTIME/Departures.aspx